

Communication Protocol of Residential & Commercial PV Grid-Connected Inverters

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Changes

Version	Date	Change Description
V1.1.73	2024-9-14	The following changes have been made based on V1.1.72: 1. Deleted the string inverters and related parameters of Utility PV Inverter BU. 2. Added the usage description of the power regulation parameters. 3. Optimized adaptive model information.
V1.1.74	2025-2-5	1.The power limit range (0.1%), power limit range, and reactive power limit range (0.1kvar) of SG150CX are modified from 0-1000, 0-1500, and -900-900 to 0-1100, 0-1650, and -990-990 respectively
V1.1.75	2025-2-27	Update "1.2 System Requirements" information.
V1.1.76	2025-6-27	Add inverter model: SG100CX-P2, SG50CX-P2-LV, Refer to Appendix 1
V1.1.77	2025-9-3	1、 Add inverter model: SG125CX-P3, Refer to Appendix 1 2、 Add Inverter fault code:681、 1328, Refer to Appendix 4
V1.1.78	2025-9-10	1、 Add inverter model: SGRS-L2, Refer to Appendix 1 2、 Add new Meter points: Meter A phase power、 Meter B phase power、 Meter C phase power、 Daily feed-in energy、 Total feed-in energy、 Daily purchased energy、 Total purchased energy; Feed-in power limit、 Feed-in power limit value、 Feed-in power limit ratio; 3、 Update "1.2 System Requirements" information; Update "3.1 Active Power Regulation";
V1.1.79	2025-12-10	1、 The inverter model of the SGRS-L2 series shall be updated, with reference to Appendix 1.
V1.1.80	2026-03-27	1、 Added SG125CX-P3 series inverter models: SG80CX-P3, SG110CX-P3. Refer to Appendix 1. 2、 Added SG110CX-P3-CN series inverter models: SG110CX-P3-CN A, SG110CX-P3-CN, SG100CX-P3-CN A, SG100CX-P3-CN. Refer to Appendix 1.

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1 Introduction

The Modbus RTU protocol is a widely used communication protocol in the industrial field, through which the inverter can communicate with other devices via the communication link (such as RS485 bus). This protocol applies to the communication between SUNGROW residential and commercial PV grid-connected inverters and third-party upper computer monitoring softwares.

Note: The third-party upper computer monitoring softwares can read the real-time running data and fault data of the inverter through this protocol. The communication should be transmitted by the RS485 interface on the inverter or the Ethernet interface of WiNet-S/WiNet-S2/Logger1000, but not by iHomeManger or other communication modules.

1.1 Abbreviations

Table 1 Abbreviations

Name	Definition
Master station	In question-and-answer communication, the party that initiates the communication becomes the master station, such as a third-party upper computer monitoring system.
Slave station	In question-and-answer communication, the party that responds passively to the command becomes the slave station, such as an inverter.
Broadcast address	The value is fixed to 0. Through the broadcast address, you can set all slave stations on the communication link execute a command.
Register address	A register address corresponds to 2 bytes of data.
UTF8	String data, the high-byte data is in the front and the low-byte data is at back. Example: Transmission order of UTF-8 data ABCD is A, B, C, D.
U16	16-bit unsigned integer, big-endian. Example: Transmission order of U16 data 0x0102 is 01, 02.
S16	16-bit signed integer, big-endian.
U32	32-bit unsigned integer; little-endian for double-word data. Big-endian for byte data. Example: Transmission order of U32 data 0x01020304 is 03, 04, 01, 02
S32	32-bit signed integer; little-endian for double-word data. Big-endian for byte data.

1.2 System Requirements

1. Communication Interface RS485

Table 2 Communication Interface Definition

Parameter	Default setting
Slave station	Inverter: 1 - 247 settable, 1 by default
Baud rate	9600 bit/s
Check bit	Null or settable

Parameter	Default setting
Data bit	8
Stop bit	1
Mode	RTU
Appliance interface	RS485-2W cable connection

2. Ethernet (optional), Modbus TCP, Default DHCP Enable, Port: 502.

3. For adapted inverter models, See [Appendix 1 Adaptive Inverter Models](#).

4. Data Requirements:

- 1) Visit all registers by subtracting 1 from the register address.

Example: if the register is 5000 –5001, visit it using address 4999 –5000.

Entering “01 04 1387 00 02 + CRC” to check the data of address 5000 –5001.

- 2) Verify type: CRC16 generates polynomial 0xA001, little-endian.

- 3) The decimal parameters are transmitted as integer after expansion.

Example: 10.333 kW is transmitted as 10333, 800.5 V is transmitted as 8005.

- 4) Negative numbers are transmitted as complement.

Example: 0xFFFF signifying -1.

- 5) Unavailable register cannot be viewed or set. The return of unsigned number is F,

Example: “0xFFFF” is the return for U16, “0xFFFFFFFF” is the return for U32;

The return of signed number is the max. positive number,

Example: “0x7FFF” for S16, “0x7FFFFFFF” for S32; 0x00 for UTF-8.

UTF-8 occupies 1 byte.

The length of odd number is complemented by 0x00.

- 6) The RW (Read-Write) registers described in section 2.2 Parameter Setting Definition do not support frequent periodic reading and setting when it is forwarded through communication modules such as WiNet-S/WiNet-S2/Logger1000.
- 7) The RW (Read-Write) registers described in section 2.2 Parameter Setting Definition are recommended to be configured during the initial communication setup and when register values change. There is no need to set them when there is no change to avoid occupying the bus.

2 Register Definition

2.1 Running Information Variable

Table 3 Running Information Variable (RO (read-only) register, Support Modbus Command 0x04)

No.	Name	Register	Data type	Data range	Unit/Ratio	Remarks
1.	Protocol No.	4950 - 4951	U32			
2.	Protocol version	4952 - 4953	U32			
3.	ARM software version	4954 - 4968	UTF-8			
4.	DSP software version	4969 - 4983	UTF-8			
5.	Reserved	4984 - 4989	U16			
6.	SN	4990 - 4999	UTF-8			Data type : UTF-8
7.	Device type code	5000	U16			See Appendix 1 Adaptive Inverter Models .
8.	Nominal active power	5001	U16		0.1kW	
9.	Output type	5002	U16	0-two phase; 1-3P4L; 2-3P3L		0: Only 5019、5022 address value is vaild; 1: 5019-5021 address is “phase x voltage” 2: 5019-5021 address is “x-x line voltage”
10.	Daily power yields	5003	U16		0.1kWh	
11.	Total power yields	5004 - 5005	U32		kWh	Compared with “Total power yields” 5144–5145, only the resolution is different.
12.	Total running time	5006 - 5007	U32		h	
13.	Internal temperature	5008	S16		0.1℃	
14.	Total apparent power	5009 - 5010	U32		VA	
15.	MPPT 1 voltage	5011	U16		0.1V	
16.	MPPT 1 current	5012	U16		0.1A	
17.	MPPT 2 voltage	5013	U16		0.1V	
18.	MPPT 2 current	5014	U16		0.1A	See Appendix 1 Adaptive Inverter Models .
19.	MPPT 3 voltage	5015	U16		0.1V	
20.	MPPT 3 current	5016	U16		0.1A	
21.	Total DC power	5017 - 5018	U32		W	
22.	A-B line voltage/phase A voltage	5019	U16		0.1V	Output type (address: 5002) is
23.	B-C line Voltage/phase B Voltage	5020	U16		0.1V	1: upload phase voltage; 2: upload line voltage
24.	C-A line Voltage/phase C	5021	U16		0.1V	Except SG5.5RS-JP

No.	Name	Register	Data type	Data range	Unit/Ratio	Remarks
	Voltage					
25.	Phase A current	5022	U16		0.1A	Except SG5.5RS-JP
26.	Phase B current	5023	U16		0.1A	
27.	Phase C current	5024	U16		0.1A	
28.	Reserved	5025 - 5030	U32			
29.	Total active power	5031 - 5032	S32		W	
30.	Total reactive power	5033 - 5034	S32		Var	
31.	Power factor	5035	S16		0.001	> 0 means leading < 0 means lagging
32.	Grid frequency	5036	U16		0.1Hz	Compared with “Grid frequency” 5148, only the resolution is different.
33.	Reserved	5037	U16			
34.	Work state 1	5038	U16	See Appendix 2 Working State 1.		Data of address 5039 - 5045 are additional
	Fault/Alarm time: Year	5039	U16			Fault/Alarm time and code (5039 - 5045) are valid only when the device work state is fault (0x5500) or alarm (0x9100)
	Fault/Alarm time: Month	5040	U16			
	Fault/Alarm time: Day	5041	U16			
	Fault/Alarm time: Hour	5042	U16			
	Fault/Alarm time: Minute	5043	U16			
	Fault/Alarm time: Second	5044	U16			Except SG5.5RS-JP
	Fault/Alarm code 1	5045	U16	See Appendix 4 Device Fault Code.		
	Reserved	5046 - 5048	U16			
35.	Nominal reactive power	5049	U16		0.1kVar	
36.	Reserved	5050 - 5070	U32			
37.	Array insulation resistance	5071	U16	1 - 20000	1kΩ	
38.	Reserved	5072-5076	U16			
39.	Active Power Regulation Setpoint	5077 - 5078	U32		1W	Except SG5.5RS-JP
40.	Reactive Power Regulation Setpoint	5079 - 5080	S32		1Var	Except SG5.5RS-JP
41.	Work state 2	5081 - 5082	U32	See Appendix 3 Working State 2.		Except SG5.5RS-JP
42.	Meter power	5083~5084	S32		1W	Only supported in the scenario where a smart energy meter is connected to the inverter for feed-in limitation.
43.	Meter A phase power	5085~5086	S32		1W	
44.	Meter B phase power	5087~5088	S32		1W	
45.	Meter C phase power	5089~5090	S32		1W	
46.	Load power	5091~5092	S32		1W	
47.	Daily feed-in energy	5093 - 5094	U32		0.1kWh	Only valid for models: SG2.0-10RS,
48.	Total feed-in energy	5095 - 5096	U32		0.1kWh	

No.	Name	Register	Data type	Data range	Unit/Ratio	Remarks
49.	Daily purchased energy	5097 - 5098	U32		0.1kWh	SGRS-L2, SG3.0-25RT, SG3.0-25RT-P2
50.	Total purchased energy	5099 - 5100	U32		0.1kWh	
51.	Reserved	5101 - 5112				
52.	Daily running time	5113	U16		min	Except SG5.5RS-JP
53.	Present country	5114	U16			
54.	MPPT 4 voltage	5115	U16		0.1V	Except SG5.5RS-JP See Appendix 1 Adaptive Inverter Models.
55.	MPPT 4 current	5116	U16		0.1A	
56.	MPPT 5 voltage	5117	U16		0.1V	
57.	MPPT 5 current	5118	U16		0.1A	
58.	MPPT 6 voltage	5119	U16		0.1V	
59.	MPPT 6 current	5120	U16		0.1A	
60.	MPPT 7 voltage	5121	U16		0.1V	
61.	MPPT 7 current	5122	U16		0.1A	
62.	MPPT 8 voltage	5123	U16		0.1V	
63.	MPPT 8 current	5124	U16		0.1A	
64.	Power factor Setpoint	5125	S16		0.001	Except SG5.5RS-JP
65.	Reserved	5126 - 5127	U16			
66.	Monthly power yields	5128 - 5129	U32		0.1kWh	Except SG5.5RS-JP
67.	MPPT 9 voltage	5130	U16		0.1V	Except SG5.5RS-JP See Appendix 1 Adaptive Inverter Models.
68.	MPPT 9 current	5131	U16		0.1A	
69.	MPPT 10 voltage	5132	U16		0.1V	
70.	MPPT 10 current	5133	U16		0.1A	
71.	MPPT 11 voltage	5134	U16		0.1V	
72.	MPPT 11 current	5135	U16		0.1A	
73.	MPPT 12 voltage	5136	U16		0.1V	
74.	MPPT 12 current	5137	U16		0.1A	
75.	Reserved	5138 - 5143	U16			
76.	Total power yields	5144 - 5145	U32		0.1kWh	Except SG5.5RS-JP
77.	Negative voltage to the ground	5146	S16	-15000 - 15000	0.1V	Except SG5.5RS-JP
78.	Bus voltage	5147	U16	0 - 15000	0.1V	
79.	Grid frequency	5148	U16		0.01Hz	
80.	Reserved	5149	U16			
81.	PID work state	5150	U16	2: PID Recover Operation 4: Anti-PID Operation 8: PID Abnormity		Refer to inverter user manual to check if it has the PID function
82.	PID alarm code	5151	U16	See Appendix 5 PID Alarm Code		

No.	Name	Register	Data type	Data range	Unit/Ratio	Remarks
83.	Reserved	5152-5193	U16			
PV String current information						
1.	String 1 current	7013	U16		0.01A	Before checking the current information of one input, please make sure the hardware supports this function. If parameter can be viewed in the LCD panel or APP software (default menu-running information), the corresponding address is readable. Except SG5.5RS-JP
2.	String 2 current	7014	U16		0.01A	
3.	String 3 current	7015	U16		0.01A	
4.	String 4 current	7016	U16		0.01A	
5.	String 5 current	7017	U16		0.01A	
6.	String 6 current	7018	U16		0.01A	
7.	String 7 current	7019	U16		0.01A	
8.	String 8 current	7020	U16		0.01A	
9.	String 9 current	7021	U16		0.01A	
10.	String 10 current	7022	U16		0.01A	
11.	String 11 current	7023	U16		0.01A	
12.	String 12 current	7024	U16		0.01A	
13.	String 13 current	7025	U16		0.01A	
14.	String 14 current	7026	U16		0.01A	
15.	String 15 current	7027	U16		0.01A	
16.	String 16 current	7028	U16		0.01A	
17.	String 17 current	7029	U16		0.01A	
18.	String 18 current	7030	U16		0.01A	
19.	String 19 current	7031	U16		0.01A	
20.	String 20 current	7032	U16		0.01A	
21.	String 21 current	7033	U16		0.01A	
22.	String 22 current	7034	U16		0.01A	
23.	String 23 current	7035	U16		0.01A	
24.	String 24 current	7036	U16		0.01A	

2.2 Parameter Setting Definition

Table 4 Parameter Setting Definition (RW (Read-Write) Register, Support Modbus Command 0x03/0x06/0x10)

No.	Name	Register	Data type	Data range	Unit/Ratio	Remarks
1.	Clock: Year	5000	U16	2000 -		Receive time synchronization setting of the monitoring system
2.	Clock: Month	5001	U16	1-12		
3.	Clock: Day	5002	U16	1-31		
4.	Clock: Hour	5003	U16	0-23		
5.	Clock: Minute	5004	U16	0-59		
6.	Clock: Second	5005	U16	0-59		
7.	Start/Stop	5006	U16	0xCF (Start) 0xCE (Stop)		
8.	Power	5007	U16	0xAA: Enable;		Available when the power

No.	Name	Register	Data type	Data range	Unit/Ratio	Remarks
	limitation switch			0x55: Disable		limitation switch (5007) is enabled
9.	Power limitation setting	5008	U16	If the inverter supports Active Power Overload (5020) and enabled, set range is 0-1000, else see Appendix 1 Adaptive Inverter Models .	0.1%	
10.	Reserved	5009	U16			
11.	Feed-in power limit	5010	U16	0xAA: Enable; 0x55: Disable		1. Available when the Feed-in power limit (5010) is enabled
12.	Feed-in power limit value	5011	U16	0- Rated active power	0.01kW	2. Only supported in the scenario where a smart energy meter is connected to the inverter for feed-in limitation. Only valid for models:
13.	Reserved	5012-5014	U16			SG2.0-10RS, SGRS-L2, SG3.0-25RT, SG3.0-25RT-P2
14.	Feed-in power limit ratio	5015	U16	0-1000	0.1%	
15.	Reserved	5016-5018	U16			
16.	PF setting	5019	S16	-1000 - -800 800 - 1000	0.001	Available when the reactive power adjustment switch (5036) is set to power factor setting valid (0xA1) > 0 means leading < 0 means lagging
17.	Active Power Overload	5020	U16	0xAA: Enable; 0x55: Disable		Enable: The inverter dispatches based on the maximum apparent power. Disable: The inverter dispatches based on the rated power. Except SG5.5RS-JP
18.	Reserved	5021 - 5032	U16			
19.	Reactive Power Ratio at Night	5033	S16	0 - 1000 0 - -1000	0.1%	Available when the reactive power generation at night (5035) is enabled Except SG5.5RS-JP
20.	Local / remote control	5034	U16	1-Remote,		1-Remote: Can only set

No.	Name	Register	Data type	Data range	Unit/Ratio	Remarks
				2-Local, 3-Local & remote		via this protocol 2-Local: Can only set via App 3-Local & remote: Can set via this protocol or App, the latest setting will take effect. Except SG5.5RS-JP
21.	Night SVG Switch	5035	U16	0xAA: Enable; 0x55: Disable		Except SG5.5RS-JP
22.	Reactive power adjustment mode	5036	U16	0x55: OFF, power factor returns to 1, reactive power percentage returns to 0; 0xA1: power factor setting valid, Reactive power percentage returns to 0; 0xA2: Reactive power percentage setting valid, power factor returns to 1; 0xA3: Enable Q(P) curve configuration; 0xA4: Enable Q(U) curve configuration		
23.	Reactive power percentage setting	5037	S16	0 - 1000 0 - -1000	0.1%	Available when the reactive power adjustment switch (5036) is set to Reactive power percentage setting valid (0xA2)
24.	Reserved	5038				
25.	Power limitation adjustment	5039	U16	See Appendix 1 Adaptive Inverter Models .	0.1kW	Available when the power limitation switch (5007) is enabled. Except SG5.5RS-JP
26.	Reactive power adjustment	5040	S16	See Appendix 1 Adaptive Inverter Models .	0.1kVar	Available when the reactive power adjustment switch (5036) is set to Reactive power percentage setting valid (0xA2). Except SG5.5RS-JP
27.	PID Recovery	5041	U16	0xAA: Enable; 0x55: Disable		Except SG5.5RS-JP
28.	Anti-PID	5042	U16	0xAA: Enable; 0x55: Disable		Only valid for SG136TX
29.	Reserved	5043 - 5047	U16			

No.	Name	Register	Data type	Data range	Unit/Ratio	Remarks
30.	Q(P) curve 1	5048-5077		See Appendix 6 Q(P) Curve 1. The reactive power adjustment switch (5036) is set to Enable Q(P) curve (0xA3)		Only valid for maintenance models
31.	Q(U) curve 1	5078-5115		See Appendix 7 Q(U) Curve 1. The reactive power adjustment switch (5036) is set to Enable Q(U) curve (0xA4)		Only valid for maintenance models
32.	Q(P) curve 2	5116-5134		See Appendix 8 Q(P) Curve 2. The reactive power adjustment switch (5036) is set to Enable Q(P) curve (0xA3)		Except SG5.5RS-JP
33.	Q(U) curve 2	5135-5154		See Appendix 9 Q(U) Curve 2. The reactive power adjustment switch (5036) is set to Enable Q(U) curve (0xA4)		Except SG5.5RS-JP

Note 1: The parameter settings for night SVG and PID will be invalid if the inverter does not support the functions. Please refer to the corresponding user manual to check if it supports the functions.

Note 2: The inverter models on sale support Q(P) curve 2 and Q(U) curve 2 by default, and the inverter models phased out now support Q(P) curve 1 and Q(U) curve 1.

3 Power Regulation Parameters

3.1 Active Power Regulation

Table 5 Active Power Regulation Parameters

No.	Name	Register	Data type	Data range	Unit/Ratio	Remarks
1.	Power limitation switch	5007	U16	0xAA: Enable 0x55: Disable		
2.	Power limitation setting	5008	U16	If the inverter supports Active Power Overload (5020) and it is enabled, the set range is 0-1000, else see Appendix 1 Adaptive Inverter Models .	0.1%	Available when the power limitation switch (5007) is enabled.
3.	Active Power Overload	5020	U16	0xAA: Enable 0x55: Disable		Enable: The inverter dispatches based on the maximum apparent power. Disable: The inverter dispatches based on the rated power.
4.	Power limitation adjustment	5039	U16	See Appendix 1 Adaptive Inverter Models .	0.1kW	Available when the power limitation switch (5007) is enabled.
5.	Feed-in power limit	5010	U16	0xAA: Enable; 0x55: Disable		Available when the Feed-in power limit (5010) is enabled
6.	Feed-in power limit value	5011	U16	0- Rated active power	0.01kW	Only supported in the scenario where a smart energy meter is connected to the inverter for feed-in limitation.
7.	Feed-in power limit ratio	5015	U16	0-1000	0.1%	Only valid for models: SG2.0-10RS, SGRS-L2, SG3.0-25RT, SG3.0-25RT-P2

Note1 : The Feed-in Limitation Function (related registers: Feed-in Limitation Ratio [5015], Feed-in Limitation Value [5011]) differs from the Power Limiting Function (related register: Active Power Limit Ratio [5008], Power limitation adjustment[5039]) in their control points for restricting inverter grid-connected power. As shown in Figure 1, Point A represents the control point of the Feed-in Limitation Function [grid connection point], while Point B represents the control point of the Power Limiting Function [inverter's AC output port]. Generally, the Feed-in Limitation Function requires the configuration of a grid gateway meter.

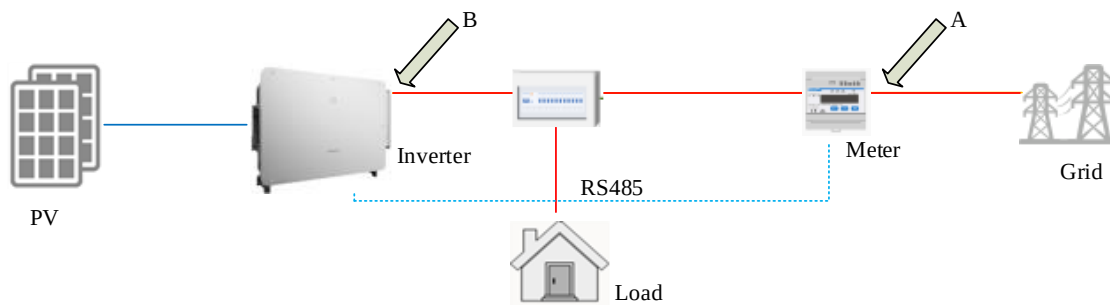


Figure 1 Schematic diagram of active power control points

3.1.1 Disable Power Limitation

The inverter will not receive any derating command. Based on the current running conditions, the maximum overload can be 110% of the rated power if the inverter supports overload running.

Method:

1. Set the power limitation switch (register 5007) to 0x55 (disabled). It is 0xAA (enabled) by default.
2. Set the Feed-in power limit (register 5010) to 0x55 (disabled).

3.1.2 Setting Power Limitation Ratio

The inverter receives the active derating command in the form of a percentage. The reference value of the percentage is the rated active power of the inverter. For inverters that support overload running, the maximum value can be set to 110%. Please refer to [Appendix 1 Adaptive Inverter Models](#) for the settable range.

For inverters that support overload running, the maximum value can be set to 100%. You can set the Active Overload Switch (register 5020) to 0xAA (enabled). In this case, the reference value of the percentage is the maximum active power of the inverter.

If the AC output terminal of the inverter is restricted (Point B in Figure 1), Method:

1. Set the power limitation switch (register 5007) to 0xAA (enabled).
2. Set the power limitation value (register 5008) to the derating target value in %.

If the point of grid connection is restricted (Point A in Figure 1), Method:

1. Set the Feed-in power limit (register 5010) to 0xAA (enabled).
2. Set the Feed-in power limit ratio (register 5015) to the derating target value in %.

3.1.3 Setting Power Limitation Value

The master inverter directly issues the target value of active power regulation. The command is an absolute power value in kW, and the adjustment accuracy is 0.1kW. The upper limit of the fixed value command is the maximum active power of the inverter, and the minimum value is 0. Please refer to [Appendix 1 Adaptive Inverter Models](#) for the settable range.

If the AC output terminal of the inverter is restricted (Point B in Figure 1), Method:

1. Set the power limitation switch (register 5007) to 0xAA (enabled).
2. Set the power limitation adjustment (register 5039) to the target value after derating in kW.

If the point of grid connection is restricted (Point A in Figure 1), Method:

1. Set the Feed-in power limit (register 5010) to 0xAA (enabled).
2. Set the Feed-in power limit value (register 5011) to the target value after derating in kW.

Note:

After the active power regulation commands are issued, the inverter will automatically maintain the command value when there is no power failure, and no periodic delivery is required. When the command value changes, the inverter will issue the new values. If the inverter is powered off and restarts, the default value is restored and the maximum power can be automatically restored. If you want to keep the set value after the inverter restarts, use the APP to set the **Active Power Setting Persistence** to **ON**.

3.2 Reactive Power Regulation**Table 6 Reactive Power Regulation Parameters**

No.	Name	Register	Data type	Data range	Unit/Ratio	Remarks
1.	PF setting	5019	S16	-1000 - -800 800 - 1000	0.001	Available when the reactive power adjustment switch (5036) is set to power factor setting valid (0xA1) > 0 means leading < 0 means lagging
2.	Reactive power adjustment mode	5036	U16	0x55: OFF, power factor returns to 1, reactive power percentage returns to 0; 0xA1: power factor setting valid, Reactive power percentage returns to 0; 0xA2: Reactive power percentage setting valid, power factor returns to 1; 0xA3: Enable Q(P) curve configuration; 0xA4: Enable Q(U) curve configuration		
3.	Reactive power percentage setting	5037	S16	0 - 1000 0 - -1000	0.1%	Available when the reactive power adjustment switch (5036) is set to Reactive power percentage setting valid (0xA2)
4.	Reactive power	5040	S16	See Appendix 1 Adaptive Inverter Models .	0.1kVar	Available when the reactive power

No.	Name	Register	Data type	Data range	Unit/Ratio	Remarks
	adjustment					adjustment switch (5036) is set to Reactive power percentage setting valid (0xA2)
5.	Q(P) curve 1	5048-5077		See Appendix 6 Q(P) Curve 1 . The reactive power adjustment switch (5036) is set to Enable Q(P) curve (0xA3)		Valid for maintenance models
6.	Q(U) curve 1	5078-5115		See Appendix 7 Q(U) Curve 1 . The reactive power adjustment switch (5036) is set to Enable Q(U) curve (0xA4)		Valid for maintenance models
7.	Q(P) curve 2	5116-5134		See Appendix 8 Q(P) Curve 2 . The reactive power adjustment switch (5036) is set to Enable Q(P) curve (0xA3)		
8.	Q(U) curve 2	5135-5154		See Appendix 9 Q(U) Curve 2 . The reactive power adjustment switch (5036) is set to Enable Q(U) curve (0xA4)		

3.2.1 Disable Reactive Power Output

The inverter will not output reactive power and the grid-connected power factor is 1.

Method:

Set the reactive power adjustment mode (register 5036) to 0x55 (disabled, by default).

3.2.2 Setting PF

The inverter receives reactive power regulation commands in the form of power factor.

Method:

1. Set the reactive power adjustment mode (register 5036) to 0xA1.
2. Set the PF (register 5019) to the target value.

3.2.3 Setting Reactive Power Ratio

The inverter receives reactive power regulation commands in the form of a percentage, and the reference value of percentage is the rated reactive power of the inverter.

Method:

1. Set the reactive power adjustment mode (register 5036) to 0xA2;
2. Set the reactive power percentage (register 5037) to the target value in %.

3.2.4 Setting Reactive Power Value

The master inverter directly issues the target value of reactive power regulation, the command is an absolute reactive power value in kVar, and the adjustment accuracy is 0.1 kVar. Please refer to [Appendix 1 Adaptive Inverter Models](#) for the settable range.

Method:

1. Set the reactive power adjustment mode (register 5036) to 0xA2.
2. Set the reactive power adjustment (register 5040) to the target value in kVar.

3.2.5 Setting the Q(P) Mode

The master inverter directly issues the target value of Q(P) curve. Curve A is in the form of power factor and curve B is in the form of reactive power ratio.

Method:

1. Set reactive power adjustment mode (register 5036) to 0xA3.
2. Take Q(P) curve 2 as an example, set Q(P) curve (registers 5116 to 5134) to the target values.

3.2.6 Setting the Q(U) Mode

The master inverter directly issues the target value of Q(P) curve in the form of reactive power ratio.

Method:

1. Set reactive power adjustment mode (register 5036) to 0xA4.
2. Take Q(U) curve 2 as an example, set Q(U) curve (registers 5135 to 5154) to the target values.

Note:

After the reactive power regulation commands are issued, the inverter will automatically maintain the command value when there is no power failure, and no periodic delivery is required. When the command value changes, the inverter will issue the new values. If the inverter is powered off and restarts, the default value is restored and the reactive power output is automatically disabled. If you want to keep the set value after the inverter restarts, use the APP to set the **Reactive Power Setting Persistence** to **ON**.

4 Examples

4.1 Acquire One Piece of Running Information

Supposed that the inverter address is 1, it needs to acquire data from address 5000 of RO (Read-Only) type.

The PC sends (HEX):

01 04 13 87 00 01 85 67

The inverter replies (HEX):

01 04 02 01 32 39 75

Note: The type code of inverter SG60KU-M is 0x0132.

4.2 Acquire Multiple Running Information

Supposed that the inverter address is 1, it needs to acquire 10 data from address starting from 5000 of (Read-Only) type.

The PC sends (HEX):

01 04 13 87 00 0A C4 A0

The inverter replies (HEX):

01 04 14 01 32 00 28 00 00 00 00 00 05 00 00 00 26 00 00 00 00 00 00 56 EA

Note: The type code of inverter SG60KU-M is 0x0132. The nominal output power is 4.0kW, two-phase. Daily power generation is 0. The total power generation is 5kWh. The total running time is 38h. The internal temperature is 0°C. The internal transformer temperature is 0°C.

4.3 Read One Setting Datum

Supposed that the inverter address is 1, it needs to read data from address 5000 of RW (Read-Write) type.

The PC sends (HEX):

01 03 13 87 00 01 30 A7

The inverter replies (HEX):

01 03 02 07 D8 BA 2E

Note: the data read out is year 2008.

4.4 Read Multiple Setting Data

Supposed that the inverter address is 1, it needs to read 10 data from address starting from 5000 of RW (Read-Write) type.

The PC sends (HEX):

01 03 13 87 00 0A 71 60

The inverter replies (HEX):

01 03 14 07 DA 00 0A 00 1E 00 09 00 28 00 25 00 CE 00 AA 01 F4 00 00 80 53

Note: The data are October, 10, 2010, 09:40:37; Stop; power limitation on, power limitation value is 50 %.

4.5 Set One Datum

Supposed that the inverter address is 1, it needs to set data from address 5000 of RW (Read-Write) type.

The PC sends (HEX):

01 10 13 87 00 01 02 07 DA 19 4D

The inverter replies (HEX):

01 10 13 87 00 01 B5 64

Or

The PC sends (HEX):

01 06 13 87 07 DA BE CC

The inverter replies (HEX):

01 06 13 87 07 DA BE CC

Note: The setting data is year 2010

4.6 Set Multiple Data

Supposed that the inverter address is 1, it needs to set 10 data to address starting from 5000 of RW (Read-Write) type.

The PC sends (HEX):

01 10 13 87 00 0A 14 07 D9 00 0A 00 1E 00 09 00 10 00 00 00 CE 00 AA 01 F4 00 00 3E 65

The inverter replies (HEX):

01 10 13 87 00 0A F4 A3

Note: The data are October, 30, 2009, 09:16:00, stop, power limitation on, power limitation value is 50 %.

4.7 Acquire SN

Supposed that the inverter address is 1, it needs to acquire 10 data from address starting from 4990 of RO (Read-Only) type.

The PC sends (HEX):

01 04 13 7D 00 0A E4 91

The inverter replies (HEX):

01 04 14 31 32 31 32 31 32 30 30 31 00 00 00 00 00 00 00 00 00 00 9B 56

Note:

1. SN data type is UTF-8.
2. Serial number is: 121212001.

4.8 Read Device Running Information

Supposed that the inverter address is 1, it needs to set 8 data to address starting from 5038 of RO (Read-Only) type.

The PC sends (HEX):

01 04 13 AD 00 0864 A9

The inverter replies (HEX):

01 04 10 55 00 07 DF 00 0C 00 15 00 04 00 0C 00 3B 00 0A EE D1

Note:

- 1) Device running state is Fault(0x5500); the fault/alarm time and code are valid in this state.
- 2) Fault time: 4 (0x0004):12(0x000C): 59(0x003B), Dec. (0x000C), 21(0x0015), 2015(0x07DF); the fault is island (0x000A).

5 Appendix

Appendix 1 Adaptive Inverter Models

Note:

1. String/MPPT means the string number in a MPPT. For example, the string/MPPT of SG36CX-P2H-CN is 2;1;1, which means that the inverter has a total of 3 MPPTs and a total of 4 strings, 2 strings to the 1st MPPT, 1 string to the 2nd MPPT, and 1 string to the 3rd MPPT.
2. If the value of string/MPPT is 1, it indicates that no string information (7013-7036) is uploaded to the read-only register.

Table 7 Three-phase Inverters (On sale)

Model series	Model	Code	MPPT	String/MPPT	Power limited range (0.1%)	Power limited range (0.1kW)	Reactive power limited range (0.1kvar)
China:							
SG3.0~25RT-20	SG3.0RT-20	0x243D	2	1	0-1100	0-33	-15 - 15
	SG4.0RT-20	0x243E	2	1	0-1100	0-44	-20 - 20
	SG5.0RT-20	0x2430	2	1	0-1100	0-55	-25 - 25
	SG6.0RT-20	0x2431	2	1	0-1100	0-66	-30 - 30
	SG7.0RT-20	0x243C	2	1	0-1100	0-77	-35 - 35
	SG8.0RT-20	0x2432	2	1	0-1100	0-88	-40 - 40
	SG10RT-20	0x2433	2	1	0-1100	0-110	-50 - 50
	SG11RT-20	0x243A	2	1	0-1100	0-121	-55 - 55
	SG12RT-20	0x2434	2	1	0-1100	0-132	-60 - 60
	SG15RT-20	0x2435	2	2;1	0-1100	0-165	-75 - 75
	SG17RT-20	0x2436	2	2	0-1100	0-187	-85 - 85
	SG20RT-20	0x2437	2	2	0-1100	0-220	-100 - 100
	SG22RT-20	0x2438	2	2	0-1100	0-242	-110 - 110
	SG23RT-20	0x243B	2	2	0-1100	0-253	-115 - 115
	SG25RT-20	0x2439	2	2	0-1100	0-275	-125 - 125
SG10~30T-CN	SG10T-CN	0x2473	2	1	0-1100	0-110	-50 - 50
	SG12T-CN	0x2474	2	1	0-1100	0-132	-60 - 60
	SG15T-CN	0x2475	2	1	0-1100	0-165	-75 - 75
	SG17T-CN	0x2476	2	2;1	0-1100	0-187	-85 - 85
	SG20T-CN	0x2477	2	2;1	0-1100	0-220	-100 - 100
	SG23T-CN	0x247B	2	2;1	0-1100	0-253	-115 - 115
	SG25T-CN	0x2479	2	2;1	0-1100	0-275	-125 - 125
	SG30T-CN	0x247C	2	2	0-1100	0-330	-150 - 150
SG20~30T-P2-CN	SG20T-P2-CN	0x2497	2	2	0-1100	0-220	-100 - 100

Model series	Model	Code	MPPT	String/MPPT	Power limited range (0.1%)	Power limited range (0.1kW)	Reactive limited range (0.1kvar)	power range
	SG23T-P2-CN	0x2498	2	2	0-1100	0-253	-115 - 115	
	SG25T-P2-CN	0x2499	2	2	0-1100	0-275	-125 - 125	
	SG30T-P2-CN	0x249A	3	2;2;1	0-1100	0-330	-150 - 150	
	SG33T-P2-CN	0x249B	3	2;2;1	0-1100	0-363	-165-165	
SG30~110CX	SG30CX	0x2C10	3	2	0-1000	0-299	-179 - 179	
	SG33CX	0x2C00	3	2	0-1100	0-363	-218 - 218	
	SG40CX	0x2C01	4	2	0-1100	0-440	-264 - 264	
	SG50CX	0x2C02	5	2	0-1100	0-550	-330 - 330	
	SG75CX	0x2C22	9	2	0-1000	0-750	-450 - 450	
	SG100CX	0x2C05	9	2	0-1100	0-1100	-660 - 660	
	SG110CX	0x2C06	9	2	0-1100	0-1100	-660 - 660	
SG30~110CX-P2-CN	SG30CX-P2-CN	0x2C23	3	2	0-1100	0-330	-198-198	
	SG33CX-P2-CN	0x2C2F	3	2	0-1100	0-363	-217.8-217.8	
	SG36CX-P2-CN	0x2C30	3	2	0-1100	0-396	-237.6-237.6	
	SG40CX-P2-CN	0x2C24	3	2	0-1100	0-440	-264-264	
	SG50CX-P2-CN	0x2C31	4	2	0-1100	0-550	-330 - 330	
	SG110CX-P2-CN	0x2C2C	9	2	0-1100	0-1210	-726 - 726	
	SG100CX-P2-CN	0x2C82	9	2	0-1100	0-1100	-660 - 660	
SG36~110CX-P2H-CN	SG36CX-P2H-CN	0x2C4A	3	2;1;1	0-1100	0-396	-237.6-237.6	
	SG40CX-P2H-CN	0x2C4B	3	2;1;1	0-1100	0-440	-264-264	
	SG50CX-P2H-CN	0x2C4C	4	2;1;1;1	0-1100	0-550	-330 - 330	
	SG110CX-P2H-CN	0x2C4D	9	2;2;1;1;1;1;1;1;1	0-1100	0-1210	-726 - 726	
SG150CX-CN	SG150CX-CN-V12	0x2C58	7	3	0-1100	0-1650	-990 - 990	

Model series	Model	Code	MPPT	String/MPPT	Power limited range (0.1%)	Power limited range (0.1kW)	Reactive limited range (0.1kvar)	power range
	SG150CX-CN-V11	0x2C59	7	3	0-1100	0-1650	-990 - 990	
SG36-50CX-P3-CN	SG36CX-P3-CN	0x249C	4	2;2;2;1	0-1100	0-396	-237.6-237.6	
	SG40CX-P3-CN	0x249D	4	2;2;2;1	0-1100	0-440	-264-264	
	SG50CX-P3-CN	0x249E	4	2	0-1100	0-550	-330-330	
SG40-136TX	SG40TX	0x2C72	6	2	0-1100	0-440	-264-264	
	SG50TX	0x2C73	6	2	0-1100			
	SG60TX	0x2C74	6	2	0-1100	0-660	-300-300	
	SG136TX	0x2C08	12	2	0-1100	0-1500	-900 - 900	
Overseas:								
SG3.0~25RT	SG3.0RT	0x243D	2	1		0-33	-15~15	
	SG4.0RT	0x243E	2	1		0-44	-20~20	
	SG5.0RT	0x2430	2	1		Default: 0-55 (Germany, Australia:0-50)	-25~25	
	SG6.0RT	0x2431	2	1		Default: 0-66 (Germany, Australia:0-60)	-30~30	
	SG7.0RT	0x243C	2	2;1		Default: 0-77 (Germany:0-70; Australia: 0-69.99)	-35~35	
	SG8.0RT	0x2432	2	2;1	Default: 0-1100 (Germany, Belgium, Australia: 0-1000)	Default: 0-88 (Germany, Australia:0-80)	-40~40	
	SG10RT	0x2433	2	2;1		Default: 0-110 (Germany, Belgium,Australia:0-100)	-50~50	
	SG12RT	0x2434	2	2;1		Default: 0-132 (Germany, Australia:0-120)	-60~60	
	SG15RT	0x2435	2	2		Default: 0-165 (Germany, Australia:0-150)	-75~75	
	SG17RT	0x2436	2	2		Default: 0-187 (Germany, Australia:0-170)	-85~85	
	SG20RT	0x2437	2	2		Default: 0-220 (Germany, Australia:0-	-100~100	

Model series	Model	Code	MPPT	String/MPPT	Power limited range (0.1%)	Power limited range (0.1kW)	Reactive limited (0.1kvar)	power range
						200)		
	SG22RT	0x2438	2	2		Default: 0-242 (Germany, Australia:0-220)	-110-110	
	SG23RT	0x243B	2	2		Default: 0-253 (Germany, Australia:0-230)	-115-115	
	SG25RT	0x2439	2	2		Default: 0-275 (Germany, Australia:0-250)	-125-125	
SG3.0~25RT (Australian)	SG3.0RT	0x2488	2	1	0-1000	0-30	-15 - 15	
	SG4.0RT	0x2489	2	1	0-1000	0-40	-20 - 20	
	SG5.0RT	0x247D	2	1	0-1000	0-50	-25 - 25	
	SG6.0RT	0x247E	2	1	0-1000	0-60	-30 - 30	
	SG7.0RT	0x2487	2	2;1	0-1000	0-70	-35 - 35	
	SG8.0RT	0x247F	2	2;1	0-1000	0-80	-40 - 40	
	SG10RT	0x2480	2	2;1	0-1000	0-100	-50 - 50	
	SG12RT	0x2481	2	2;1	0-1000	0-120	-60 - 60	
	SG15RT	0x2482	2	2;2	0-1000	0-150	-75 - 75	
	SG17RT	0x2483	2	2;2	0-1000	0-170	-85 - 85	
	SG20RT	0x2484	2	2;2	0-1000	0-200	-100 - 100	
	SG23RT	0x2486	2	2;2	0-1000	0-230	-115 - 115	
	SG25RT	0x2485	2	2;2	0-1000	0-250	-125 - 125	
SG3.0~25RT-P2	SG3.0RT-P2	0x244D	2	1	Default: 0-1100 (Germany, Belgium, Australia: 0-1000)	Default: 0-33 (Germany, Australia:0-30)	-15~15	
	SG4.0RT-P2	0x244E	2	1		Default: 0-44 (Germany, Australia:0-40)	-20~20	
	SG5.0RT-P2	0x2440	2	1		Default: 0-55 (Germany, Australia:0-50)	-25~25	
	SG6.0RT-P2	0x2441	2	1		Default: 0-66 (Germany, Australia:0-60)	-30~30	
	SG7.0RT-P2	0x244C	2	2;1		Default: 0-77 (Germany:0-70; Australia: 0-69.99)	-35~35	
	SG8.0RT-P2	0x2442	2	2;1		Default: 0-88 (Germany, Australia:0-80)	-40~40	
	SG10RT-P2	0x2443	2	2;1		Default: 0-110	-50~50	

Model series	Model	Code	MPPT	String/MPPT	Power limited range (0.1%)	Power limited range (0.1kW)	Reactive limited range (0.1kvar)	power range
						(Germany, Australia:0-100)		
	SG12RT-P2	0x2444	2	2;1		Default: 0-132 (Germany, Australia:0-120)	-60~60	
	SG15RT-P2	0x2445	2	2		Default: 0-165 (Germany, Australia:0-150)	-75~75	
	SG17RT-P2	0x2446	2	2		Default: 0-187 (Germany, Australia:0-170)	-85~85	
	SG20RT-P2	0x2447	2	2		Default: 0-220 (Germany, Australia:0-200)	-100~100	
	SG23RT-P2	0x244B	2	2		Default: 0-253 (Germany, Australia:0-230)	-115~115	
	SG25RT-P2	0x2449	2	2		Default: 0-275 (Germany, Australia:0-250)	-125~125	
SG25~110CX	SG25CX-SA	0x2C15	3	2	0-1100	0-275	-165~165	
	SG30CX	0x2C10	3	2	0-1000	0-299	-179~179	
	SG33CX	0x2C00	3	2	0-1100	0-363	-218~218	
	SG40CX	0x2C01	4	2	0-1100	0-440	-264~264	
	SG50CX	0x2C02	5	2	0-1100	0-550	-330~330	
	SG75CX	0x2C22	9	2	0-1000	0-750	-450~450	
	SG110CX	0x2C06	9	2	0-1100	0-1100	-660~660	
	SG49.5CX-JP	0x2C21	6	2	0-1000	0-495	-297 ~ 297	
SG36~60CX-US	SG100CX-JP	0x2C12	12	2	0-1000	0-1000	-660~660	
	SG36CX-US	0x2C0A	4	2	0-1100	0-360	-216~216	
SG25~125CX-P2	SG60CX-US	0x2C0B	6	2	0-1100	0-600	-360~360	
	SG25CX-P2	0x2C33	3	2	0-1100	0-275	-165~165	
	SG30CX-P2	0x2C34	3	2	0-1100	0-330	-198~198	
	SG33CX-P2	0x2C35	3	2	0-1100	0-363	-217~217	
	SG36CX-P2	0x2C36	4	2	0-1100	0-396	-237~237	
	SG40CX-P2	0x2C37	4	2	0-1100	0-440	-264~264	
	SG50CX-P2	0x2C32	4	2	0-1100	0-550	-330~330	
	SG75CX-P2	0x2C3F	8	2	0-750	0-750	-450~450	
	SG110CX-P2	0x2C40	12	2	0-1100	0-1100	-660~660	
	SG125CX-P2	0x2C2D	12	2	0-1100	0-1250	-750 - 750	

Model series	Model	Code	MPPT	String/MPPT	Power limited range (0.1%)	Power limited range (0.1kW)	Reactive limited (0.1kvar)	power range
	SG50CX-P2-JP	0x2C60	4	2	0-1000	0-500	-330 - 330	
	SG100CX-P2	0x2C7F	12	2	0-1000	0-1000	-600 - 600	
SG15-75CX-LV	SG15CX-P2-LV	0x2C53	2	4	0-1100	0-165	-99-99	
	SG20CX-P2-LV	0x2C54	3	6	0-1100	0-220	-132-132	
	SG25CX-P2-LV	0x2C55	3	6	0-1100	0-275	-165-165	
	SG75CX-LV	0x2C75	7	3	0-1000	0-750	-450-450	
	SG50CX-P2-LV	0x2C80	8	2	0-1100	0-550	-330 - 330	
	SG150CX	SG150CX	0x2C57	7	3	0-1100	0-1650	-990 - 990
SG125CX-P3	SG125CX-P3	0x2C7C	10	2	0-1000	0-1250	-750 - 750	
	SG80CX-P3	0x2C89	10	2	0-1100	0-880	-528 - 528	
	SG110CX-P3	0x2C8A	10	2	0-1100	0-1100	-660 - 660	
SG110CX-P3-CN	SG110CX-P3-CN A	0x2C7D	8	2	0-1100	0-1210	-726 - 726	
	SG110CX-P3-CN	0x2C7E	8	2	0-1100	0-1210	-726 - 726	
	SG100CX-P3-CN A	0x2C85	8	2	0-1100	0-1100	-660 - 660	
	SG100CX-P3-CN	0x2C86	8	2	0-1100	0-1100	-660 - 660	

Table 8 Three-phase Inverters (Maintenance)

Model series	Model	Code	MPPT	String/MPPT	Power limited range (0.1%)	Power limited range (0.1kW)	Reactive power limited range (0.1kvar)
China / Overseas:							
SG10-80KTL	SG10KTL	0x26	2	3	0-1100	0-110	-50-50
	SG12KTL	0x29	2	3	0-1100	0-132	-60-60
	SG15KTL	0x28	2	3	0-1100	0-165	-75-75
	SG20KTL	0x2A	2	3	0-1100	0-220	-100-100
	SG30KTL	0x27	2	4	0-1100	0-330	-150-150
	SG36KTL	0x2D	2	5	0-1100	0-396	-180-180
	SG40KTL	0x2F	2	4	0-1100	0-396	-180-180
	SG50KTL	0x0116	3	3	0-1100	0-530	-240-240
	SG60KTL	0x010F	1	14	0-1100	0-660	-300-300
	SG80KTL	0x0138	1	18	0-1000	0-800	-400-400

Model series	Model	Code	MPPT	String/MPPT	Power limited range (0.1%)	Power limited range (0.1kW)	Reactive power limited range (0.1kvar)
	SG80KTL-20	0x0138	1	18	0-1000	0-800	-400-400
	SG8KTL-M	0x013F	2	1	0-1100	0-88	-40-40
	SG10KTL-M	0x013E	2	1	Default: 0-1100 (Germany, 0-1000)	Default: 0-110 (Germany, 0-100)	-50-50
	SG12KTL-M	0x013C	2	2	0-1100	0-132	-60-60
	SG15KTL-M	0x0142	2	2	0-1100	0-165	-75-75
	SG17KTL-M	0x0149	2	2	0-1100	0-187	-85-85
	SG20KTL-M	0x0143	2	2	0-1100	0-220	-100-100
	SG30KTL-M	0x0141	3	3;3;2	0-1000	0-300	-150-150
SG8-80KTL-M	SG30KTL-M-V31	0x70	3	3;3;2	0-1000	0-300	-150-150
	SG33KTL-M	0x0134	3	3;3;2	0-1100	0-363	-165-165
	SG36KTL-M	0x74	3	3;3;2	0-1000	0-360	-180-180
	SG40KTL-M	0x0135	3	3	0-1100	0-440	-200-200
	SG50KTL-M	0x011B	4	3	0-1100	0-550	-250-250
	SG50KTL-M-20	0x011B	4	3	0-1100	0-550	-250-250
	SG60KTL-M	0x0131	4	4	0-1100	0-660	-300-300
	SG80KTL-M	0x0139	4	4	0-1100	0-880	-400-400
SG5-10KTL-MT	SG5KTL-MT	0x0147	2	1	0-1100	0-55	-25-25
	SG6KTL-MT	0x0148	2	1	0-1100	0-66	-30-30
	SG10KTL-MT	0x2C0F	2	2	0-1100	0-110	-50-50
Overseas:							
	SG30KU	0x2C	2	5	0-1100	0-330	-150-150
	SG36KU	0x2E	2	5	0-1100	0-396	-180-180
SG30~60KU	SG60KU	0x0136	1	8	0-1100	0-660	-300-300
	SG60KU-M	0x0132	4	4	0-1100	0-660	-300-300
	SG33K3J	0x013D	3	3	0-1000	0-333	-166-166
	SG49K5J	0x0137	4	3	0-1000	0-495	-247-247
Others	SG34KJ	0x72	2	4	0-1000	0-340	-170-170
	LP_P34KSG	0x73	1	4	0-1000	0-340	-170-170

Table 9 Single-phase Inverters (On sale)

Model series	Model	Code	MPPT	String/MPPT	Power limited range (0.1%)	Power limited range (0.1kW)	Reactive power limited range (0.1kvar)
China:							
SG6.0~10RS-CN	SG6.0RS-CN	0x2607	2	1	0-1100	0-66	-39.6-39.6
CN	SG8.0RS-CN	0x2608	3	1	0-1100	0-88	-52.8-52.8

Model series	Model	Code	MPPT	String/MPPT	Power limited range (0.1%)	Power limited range (0.1kW)	Reactive power limited range (0.1kvar)
	SG10RS-CN	0x2609	3	1	0-1100	0-110	-66~66
Overseas:							
SG2.0~3.0RS-S	SG2.0RS-S	0x2600	1	1	0-1000	0-20	-12~12
	SG2.5RS-S	0x2601	1	1	0-1000	0-25	-15~15
	SG3.0RS-S	0x2602	1	1	0-1000	0-30	-18~18
SG3.0~10RS	SG3.0RS	0x2603	2	1	0-1000	0-30	-18~18
	SG3.6RS	0x2604	2	1	0-1000	0-36	-21~21
	SG4.0RS	0x2605	2	1	0-1000	0-40	-24~24
	SG5.0RS	0x2606	2	1	0-1000	0-50	-30~30
	SG6.0RS	0x2607	2	1	0-1000	0-60	-36~36
	SG8.0RS	0x2608	3	1	0-1000	0-80	-48~48
	SG9.0RS	0x260E	3	1	0-1000	0-90	-54~54
	SG10RS	0x2609	3	1	0-1000	0-100	-60~60
	SG5.0RS-ADA	0x260F	3	1	0-1000	0-50	-30~30
	SG3.0RS-L	0x2612	1	1	0-1000	0-30	-18~18
SG3.0~10RS-L	SG3.3RS-L	0x2618	1	1	0-1000	0-33	-20~20
	SG4.0RS-L	0x2613	2	1	0-1000	0-40	-24~24
	SG4.6RS-L	0x2619	2	1	0-1000	0-46	-30~30
	SG5.0RS-L	0x2614	2	1	0-1000	0-50	-30~30
	SG6.0RS-L	0x2615	2	1	0-1000	0-60	-36~36
	SG7.5RS-L	0x261A	2	2,1	0-1000	0-75	-45~45
	SG8.0RS-L	0x2611	2	2,1	0-1000	0-80	-48~48
	SG9.0RS-L	0x2616	3	1	0-1000	0-90	-54~54
	SG10RS-L	0x2617	3	1	0-1000	0-100	-60~60
SG5.5RS-JP	SG5.5RS-JP	0x260D	3	3	0-1000	0-55	-35 - 35
SGRS-L2	SG3.0RS-L2	0x2622	1	1	0-1100	0-30	-18~18
	SG4.0RS-L2	0x2623	2	1	0-1100	0-40	-24~24
	SG5.0RS-L2	0x2624	2	1	0-1100	0-50	-30~30
	SG6.0RS-L2	0x2625	2	1	0-1100	0-60	-36~36
	SG7.5RS-L2	0x2626	3	1	0-1100	0-75	-45~45
	SG8.0RS-L2	0x2627	3	1	0-1100	0-80	-48~48
	SG10RS-L2	0x2628	3	1	0-1100	0-100	-60~60
	SG3.0RS-L2-IN	0x2629	1	1	0-1100	0-30	-18~18
	SG3.3RS-L2-IN	0x262A	1	1	0-1100	0-33	-20~20
	SG3.0RS-L2-S	0x262C	1	1	0-1100	0-30	-18~18
	SG3.3RS-L2-S	0x262D	1	1	0-1100	0-33	-20~20
	SG4.0RS-L2-S	0x262E	1	1	0-1100	0-40	-24~24

Appendix 2 Working State 1

Table 10 Working State (Register 5038)

State	Value read by register 5038	Paraphrase	Grid-connected power generation
Run	0x0	After being energized, inverter tracks the PV arrays' maximum power point (MPP) and converts the DC power into AC power. This is the normal operation mode.	Yes
Stop	0x8000	Inverter is stopped.	No
Key stop	0x1300	Inverter will stop operation by manually "stop" via app. In this way, inverter internal DSP stops. To restart the inverter, manually start via app.	No
Emergency Stop	0x1500	When the dry contact signal of the inverter is triggered by external device, the inverter performs an emergency shutdown and the inverter is in stop state. This function is supported only when the inverter has dry contact terminal in hardware.	No
Standby	0x1400	Inverter enters standby mode when DC side input is insufficient. In this mode inverter will wait within the standby duration.	No
Initial standby	0x1200	The inverter is in the initial power-on standby state.	No
Starting	0x1600	The inverter is initializing and synchronizing with the grid	No
Alarm run	0x9100	Warning information is detected.	Yes
Derating run	0x8100	The inverter derates actively due to environmental factors such as temperature or altitude.	Yes
Dispatch run	0x8200	The inverter runs according to the scheduling instructions received from the monitoring background.	Yes
Fault	0x5500	If a fault occurs, inverter will automatically stop operation, and disconnect the AC relay. The fault information will be displayed in the app. Once the fault is removed in recovery time, inverter will automatically resume running.	No
Communicate fault	0x2500	The internal communication between the ARM and DSP of the inverter fails, and the ARM cannot obtain the status of the DSP running information.	Unconfirmed
Uninitialized	0x1111	The inverter is not initialized.	No

Appendix 3 Working State 2

Note: The definition corresponding to the state is the same as that in Appendix 2

Table 11 Working State (Register 5081-5082)

State	Corresponding Bit in register 5081-5082	Notes
Run	0	Total run state bit BIT17
Stop	1	
Initial standby	2	
Key stop	3	
Standby	4	
Emergency Stop	5	
Starting	6	
Fault	9	Total fault state bit BIT18
Alarm run	10	Total run state bit BIT17
Derating run	11	Total run state bit BIT17
Dispatch run	12	Total run state bit BIT17
Communicate fault	13	Total fault state bit BIT18
Total run bit	17	Device is grid-connected running
Total fault bit	18	Device is in fault stop state

Appendix 4 Device Fault Code

Note: Please refer to the product user manual for handling measures.

Table 12 Fault codes

Fault code	Fault name
2, 3, 14, 15	Grid Overvoltage
4, 5	Grid Undervoltage
8	Grid Overfrequency
9	Grid Underfrequency
10	Grid Power Outage
12	Excess Leakage Current
13	Grid Abnormal
17	Grid Voltage Imbalance
28, 29, 208, 448–479	PV Reserve Connection Fault
532–547, 564–579	PV Reverse Connection Alarm
548–563, 580–595	PV Abnormal Alarm
37	Excessively High Ambient Temperature
43	Excessively Low Ambient Temperature
39	Low System Insulation Resistance
106	Grounding Cable Fault
88	Electric Arc Fault
84	Reverse Connection Alarm of the Meter/CT
514	Meter Communication Abnormal Alarm
323	Grid Confrontation
75	Inverter Parallel Communication Alarm
7, 11, 16, 19–25, 30–34, 36, 38, 40–42, 44–50, 52–58, 60–68, 85, 87, 92, 93, 100–105, 107–114, 116–124, 200–211, 248–255, 300–322, 324–326, 401–412, 600–603, 605, 608, 612, 616, 620, 622–624, 681, 800, 802, 804, 807, 1096–1122	System Fault
59, 70–72, 74, 76, 82, 83, 89, 77–81, 216–218, 220–231, 432–434, 500–513, 515–518, 900, 901, 910, 911, 635, 636, 637, 638, 86, 396–397, 1124–1127	System Alarm
264–283	MPPT Reverse Connection
332–363	Boost Capacitor Overvoltage Alarm
364–395	Boost Capacitor Overvoltage Fault
1548–1579	String Current Reflux
1600–1611	PV Grounding Fault
1616	System Hardware Fault
1328	PV cable-to-ground short circuit

Appendix 5 PID Alarm Code

Note: Please refer to the product user manual for handling measures.

Fault code	Fault name
432	PID impedance abnormality
433	PID function abnormality
434	PID overvoltage/overcurrent protection

Appendix 6 Q(P) Curve 1

5048~5064, Not Italy curve					
Curve	5048	U16	0 - 1, 0 Curve A, 1 Curve B		Curve A: $PA < PC$
PB	5049	U16	200-1000, Curve B is valid	0.1%	Curve B:
KB	5050	S16	-500-500 Curve B is valid	0.001	$(PA \leq PB \leq PC, \text{ but } PA \neq PC)$
PA	5051	U16	Curve A: 0~500 Curve B: 100-1000	0.1%	and
PC	5052	U16	Curve A: 500~1000 Curve B: 200-1000	0.1%	$(KA \leq KB \leq KC$
KA	5053	S16	Curve A: 900~1000 Curve B: -500-500	0.001	or
KC	5054	S16	Curve A: 900~1000 Curve B: -500-500	0.001	$KA \geq KB \geq KC, \text{ but } KA \neq KC)$
Reserved	5055-5064				
5065~5077, Italy curve					
powerA	5065	U16	200~1000	0.1%	$powerA \leq powerB < powerC$
powerB	5066	U16	200~1000	0.1%	
powerC	5067	U16	200~1000	0.1%	
PF_max	5068	U16	900~1000	0.001	
Uin	5069	U16	1000~1100	0.1%	$U_{in} > U_{out}$
Uout	5070	U16	900~1000	0.1%	
Reserved	5071~5077	U16			

Appendix 7 Q(U) Curve 1

5078~5097, Not Italy curve					
Curve A Curve B					
Curve	5078	U16	0-1, 0 Curve A, 1 Curve B		Curve A: (U1Limit+Hysteresis<U2 Limit-Hysteresis) and (-Upper Q/Sn<=Q_U1<= Lower Q/Sn) and (-Upper Q/Sn<=Q_U2<= Lower Q/Sn) Curve: U1 Limit == U2 Limit
Q_U1	5079	S16	-500~500, Curve A is valid	0.1%	
Q_U2	5080	S16	-500~500, Curve A is valid	0.1%	
Lower ULimit	5081	U16	800~1000	0.1%	
Upper U Limit	5082	U16	1000~1200	0.1%	
U1 Limit	5083	U16	900~1100	0.1%	
U2 Limit	5084	U16	900~1100	0.1%	
Hysteresis	5085	U16	0~50, Curve A is valid	0.1%	
Lower Q/Sn	5086	U16	(Ind) 0~ 500	0.1%	
Upper Q/Sn	5087	U16	(Cap) 0~500	0.1%	
Reserved	5088~5097				
5098~5115, Italy curve					
A: Default B					
V1i	5098	U16	900~1100	0.1%	V2i < V1i < V1s < V2s
V2i	5099	U16	900~1100	0.1%	
V1s	5100	U16	900~1100	0.1%	
V2s	5101	U16	900~1100	0.1%	
Qmax	5102	U16	500~1000	0.001	
Pin	5103	U16	200~1000	0.1%	Pin > Pout
Pout	5104	U16	10~200	0.1%	
Curve	5105	U16	0-1, 0 Curve A, 1 Curve B		
Reserved	5106~5115				

Appendix 8 Q(P) Curve 2

Curve	5116	U16	0-1 0: Curve A 1: Curve B		1. $QP_P1 \leq QP_P2 < QP_P3$ 2. If $QP_P1 = QP_P2$, $QP_K1 = QP_K2$
QP_P1	5117	U16	100~1000	0.1%	QP_EnableMode: Enter and exit Q (P) mode conditions, 0xAA, unconditional entry and exit; 0x55, conditional entry and exit: the voltage is higher than $QP_EnterVoltageRatio$, the voltage is lower than $U_{outQP_ExitVoltageRatio}$, and the power is lower than $QP_ExitPowerRatio$
QP_P2	5118	U16	200~1000	0.1%	
QP_P3	5119	U16	200~1000	0.1%	
QP_K1	5120	S16	Curve A: 800~1000 Curve B: -600~600	0.001	
QP_K2	5121	S16	Curve A: 800~1000 Curve B: -600~600	0.001	
QP_K3	5122	S16	Curve A: 800~1000 Curve B: -600~600	0.001	
QP_EnterVoltageRatio	5123	U16	1000~1100	0.1%	
QP_ExitVoltageRatio	5124	U16	900~1000	0.1%	
QP_ExitPowerRatio	5125	U16	10~200	0.1%	
QP_EnableMode	5126	U16	0xAA Yes 0x55 No		
Reserved	5127~5134	U16			

Appendix 9 Q(U) Curve 2

